

Pengping Tan

tanpp5@mail2.sysu.edu.cn | pengping-tan.github.io

School of Computer Science and Engineering, Sun Yat-sen University, Guangzhou, China

EDUCATION

Sun Yat-sen University

M.Eng. in Computer Technology

School of Computer Science and Engineering

Guangzhou, China

Sept. 2024 – Jun. 2027 (expected)

Sun Yat-sen University

B.S. in Information and Computing Science

GPA: 3.8/4.0 | Minor in Psychology

Zhuhai, China

Sept. 2020 – Jun. 2024

RESEARCH INTERESTS

Human-Computer Interaction, Human-Centered AI, Social Computing, AI-Mediated Decision-Making, Sensemaking, Reflection, and Identity Work, HumanAI Collaboration, Narrative and Storytelling, Health and Well-being Technologies

RESEARCH PROJECTS

JobMate: AI-Mediated Career Exploration with Peer Stories

Project Lead / Advisor: Prof. Zhenhui Peng

Guangzhou, China

Oct. 2025 – Apr. 2026

- Initiated and led JobMate, a research probe that turns social media peer career posts into conversational personas, to investigate how dialogue changes the way users connect others' experiences to their own situations.
- Conducted formative research (survey: $N = 64$; interviews: $N = 8$) and used the findings to design structured persona cards, keep original posts visible alongside conversations, and support SDT-informed, retrieval-augmented dialogue; independently designed the data-processing pipeline and developed the full-stack system.
- Designed and conducted a between-subjects mixed-methods study ($N = 24$) comparing JobMate with native RedNote browsing, and led the statistical and thematic analyses and manuscript preparation. Found that conversations prompted users to articulate previously implicit constraints and to accept, challenge, or revise the agent's interpretations of their situations, making the sensemaking process more explicit and open to revision. The resulting paper is under review at ACM CHI 2027.

Investor Sentiment Analysis and Stock Price Prediction with Integrated Emojis

Team Lead / Faculty Mentor: Prof. Patrick Houlihan, Columbia University

Remote

Jul. 2025 – Sept. 2025

- Led a four-person research team investigating whether emoji features could improve sentiment-aware stock-price prediction; developed the research plan, delegated tasks, and delivered biweekly progress presentations.
- Co-designed an LSTM-based multimodal model integrating FinBERT text embeddings, Emoji2Vec representations, and historical market indicators to predict next-day closing prices for three stocks using 10,000 StockTwits posts from the StockEmotions dataset.
- Led the literature review and manuscript preparation; adding emoji features reduced MSE across all three stocks, with a 42.3% reduction for Boeing.

Bai Juyi's Poetry and *Baishi Liutie*: A Computational Study

Undergraduate Thesis / Advisors: Prof. Baoquan Zhao & Prof. Zhitian Wang

Zhuhai, China

Nov. 2023 – May 2024

- Proposed and independently implemented a computational extension of Prof. Wang's manual study to identify textual connections between Bai Juyi's poetry and *Baishi Liutie*.
- Built a Python pipeline using rule-based text processing and exact matching to analyze 40 poetry volumes and 20,667 terms and allusions, identifying 1,246 matches and broadly reproducing the manual study's distributional trends.

PUBLICATIONS

- Tan, P., Zhao, B., Ma, S., & Peng, Z. (2026). Same Stories, Different Journeys: Exploring Persona-Grounded Conversational Agents for Supporting Career Exploration with Peers' Posts. **Under review at ACM CHI 2027.** <https://arxiv.org/abs/2607.11039>

TEACHING EXPERIENCE

Teaching Assistant, Sun Yat-sen University

Mental Health Education: Coordinate classes, communicate with students, and grade assignments.

Sep. 2026 – Present

Private Mathematics Tutor, China

Tutored nine students in weekly two-hour individualized high school mathematics sessions.

Jan. 2024 – Jul. 2024

AWARDS & SCHOLARSHIPS

- Outstanding Student Scholarship, Second Prize, Sun Yat-sen University 2023 – 2024
- Outstanding Student Scholarship, Third Prize, Sun Yat-sen University 2021 – 2022; 2022 – 2023

SKILLS

- Research Methods:** Surveys, semi-structured interviews, mixed-methods user studies, statistical and thematic analysis
- Technical Skills:** Python, PyTorch, JavaScript, LLM APIs, Vue.js, Node.js, MongoDB, Neo4j
- Languages:** English (IELTS Academic: 7.0), Mandarin Chinese (native)